

SACHIN KURUP

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RESEARCH FOCUS

I am interested in autonomous language-based systems: how LLM agents reason and decide in complex environments, use external tools safely, resist adversarial manipulation, and judge when to act alone versus defer to human oversight. I have worked on this from both sides, building multi-agent systems with guardrailed tool access and decision-adherence checks, and studying failures empirically in my first-author work *When Retrieval Hurts* (MeLLMs @ ACL 2026). I am pursuing doctoral study to make agent reasoning reliable enough to support discovery in complex scientific and technical domains.

RESEARCH INTERESTS

Autonomous LLM agents and decision-making in complex environments · Safe tool use and adversarial robustness · Human oversight and deferral in agentic systems · Reliable reasoning for scientific and technical discovery · Grounding and evaluation of agentic and retrieval systems

EDUCATION

B.Tech, Computer Science & Engineering - Amrita Vishwa Vidyapeetham, Kerala, India

Aug 2026

- CGPA: 8.65/10.0
- Selected coursework: Artificial Intelligence, Machine Learning, Distributed Systems, Algorithm Design, Quantum Computing

PUBLICATIONS

PEER-REVIEWED

S. Kurup, G. Menon, Varalekshmy, Swathi, V. Kangirangat, and Veena. “When Retrieval Hurts: Evidence Utilization, Script Fidelity, and Knowledge Conflicts in Multilingual RAG.” Workshop on Multilingual Large Language Models (MeLLMs), ACL 2026.

S. Kurup, G. Menon, Varalekshmy, Swathi, V. Kangirangat, and Veena. “Scope Matters: A Hallucination-Aware Evaluation of Retrieval Strategies for Multilingual RAG.” CLEF 2026.

OTHER DOMAINS

S. Kurup, Lakshmi K, Arunav H, Venkata Sai Teja Sunku, Chejarla Venkata Surya Sai Anish, Dhivvy J P, Renjith Mohan. “Digital Twin Framework for Precision Mushroom Cultivation Using IoT Sensors.” Proc. Intl. Conf. on Sustainable & Resilient Futures (ICSRF), 2025. (Published) Patent pending.

P. Rohit, A. Singh, V. Mageshen, K. Sivasabari, S. Kurup. “Hyperspectral Analysis for Magnesium Estimation in Soil.” IEEE InGARSS 2024, pp. 1-4. doi:10.1109/InGARSS61818.2024.10984012. (Published)

S. Kurup, Lakshmi K., and Renjith Mohan. “Advanced Control Systems for Digital Twin-Based Mushroom Cultivation: Part II.” In preparation, 2026.

PATENTS FILED

- Scientific Digital Twin Framework for Biological State Estimation and Synchronization
- Biological-State-Aware Autonomous Environmental Control System for Controlled Mushroom Cultivation
- Computer Vision-Based Mushroom Phenotyping and Biological Health Assessment System

RESEARCH & ENGINEERING EXPERIENCE

Quantum Project Engineer - Center for Cybersecurity Systems and Networks, Amrita Vishwa Vidyapeetham

Oct 2026 - Present

Development of Quantum Error Correction Appliance for Superconducting Quantum Computers using Rudra Server - MeitY-funded

- Error correction on superconducting qubits is hard to keep optimized as device noise shifts; on this project, funded by MeitY (Ministry of Electronics and Information Technology), I am building an AI pipeline to optimize error correction.
- The pipeline takes syndrome data from the readout and extraction stages and runs noise estimation, syndrome processing, an adaptive decoder, error prediction, and correction optimization, passing correction decisions back to the control electronics.

Research Intern - Prevalence Health (Embrace Ventures), California, USA

Jul 2026 - Present

- Clinics running structured studies often lack the infrastructure to operationalise protocols consistently at scale; I am building agentic workflows and systems that help translate clinical study protocols into reliable, repeatable clinic operations, working directly with the organisation's clinical and research leadership.
- Collaborating closely with researchers and clinicians trained at Stanford, MIT, and Harvard, contributing to the design of agent-driven systems intended to reduce protocol-deviation errors and free up clinical staff time for patient care.

Software Engineering Intern - SeedlingLabs Pvt. Ltd., Bangalore, India

Jan 2026 - Jun 2026

Agentic memory and decision-grounding - internal production system

- Agents across internal tools were reasoning over siloed, inconsistent knowledge, causing contradictory outputs; I built a multi-agent memory layer integrating these applications into a shared organisational knowledge graph, giving agents consistent context to query and reason over.
- To catch agents silently drifting from approved actions, I developed a decision-adherence component linking operational actions to prior decisions and flagging divergences, giving whoever owns agent oversight a concrete trust and accountability signal.
- Uncontrolled write access across agent services risked silent state corruption; I designed MCP servers with CRUD guardrails for validation and access control, giving downstream teams a reliable, auditable communication backbone.

Agentic educational platform - institutional deployment

- Manual grading, question-paper creation, and lesson planning consume significant instructor time at scale; I co-developed agentic pipelines for automated answer-script evaluation, question-paper generation, and personalised lesson planning, deployed across partner institutions, reducing manual workload for instructors and administrators (system design, prompting, and backend integration).

AI planning for cyber-physical systems security - patent pending

- Critical-infrastructure defenders currently identify attack paths largely by hand, an approach that doesn't scale with system complexity; I developed an AI-planning system that analyses attack-connection graphs and fault trees to identify vulnerability paths in cyber-physical systems, with a human-in-the-loop process for reasoning over CVE-based exploits, giving security engineers interpretable, actionable results across system layers.

Research Intern, Computational Cardiology - Health and AI Lab, Amrita University, Kerala, India

Jan - Aug 2025

- Chagas disease frequently goes undiagnosed in under-resourced, endemic regions of South America, where late detection worsens outcomes; I applied transformer architectures and prompt engineering to ECG-based classification, benchmarking hybrid sequence models (best from an LSTM-Transformer) for the PhysioNet 2025 Challenge, and analysed epidemiological risk patterns across affected populations, contributing toward automated screening tools clinicians in these regions could rely on.

SELECTED PROJECTS

When Retrieval Hurts: Multilingual RAG Analysis

Jan 2026 - Apr 2026

- RAG practitioners deploying systems for non-English, non-Latin-script users currently have no reliable signal for when retrieval will backfire; I conducted an empirical study of RAG across 21 typologically diverse languages and 5 LLMs, comparing RAG against a non-RAG baseline across five prompting strategies and multiple retrieval configurations. Found that high retrieval quality does not guarantee performance gains - models consistently underutilise retrieved evidence - and identified script fidelity as a key factor behind hallucination in non-Latin-script languages. Introduced lightweight inference-time metrics that let RAG builders and researchers detect these failure modes directly. (Python, PyTorch, Hugging Face Transformers, FAISS, LangChain, ONNX Runtime)

Multi-Agent Retrieval-Augmented QA Framework

Jul 2025 - Dec 2025

- Single-pass RAG pipelines can't self-correct, so errors compound silently in production; I built a multi-agent system coordinating retrieval, reasoning, and generation, with an autonomous judge and reinforcement-learning feedback to iteratively improve answer quality, useful for teams building agentic QA systems where answers can't be manually checked at scale. (LangChain, LlamaIndex, Hugging Face Transformers, FAISS)

Cyber-Physical Systems Security Platform - Colorado State University, with Prof. Indrajit Ray

Jun 2025 - Jun 2026

- Security researchers typically hand-build attack-defense test environments, which is slow and hard to reproduce; I designed and implemented a network orchestration controller managing the full VM lifecycle (topology, networking, live access), giving researchers a reproducible, on-demand platform to run attack-defense scenarios programmatically.
- Separately, to help defenders prioritise patches instead of manually tracing fault paths by hand, I developed resiliency graph analysis for cyber-physical systems: converting fault-tree data into PDDL and modelling fault propagation for AI-planning-based vulnerability path identification. (Python, Flask, VBoxManage, paramiko, NetworkX, PDDL)

Additional projects: quantum search for constrained word-guessing (Grover's algorithm, Qiskit); RL-based scheduler for distributed data centres; LLM-based personalised e-learning platform; digital-twin mushroom cultivation (patent pending); ML for soil-nutrient estimation from hyperspectral imaging (published); XSS-vulnerability classification.

TECHNICAL SKILLS

- Languages: Python, C++, C, TypeScript, SQL
- LLM / NLP / Agents: multi-agent architectures, MCP servers and tool-use guardrails, RAG systems, LLM evaluation, knowledge graphs, prompt engineering, LLM fine-tuning, reinforcement learning, AI planning (PDDL); LangChain, LlamaIndex, Hugging Face Transformers, FAISS
- ML: PyTorch, TensorFlow, scikit-learn, Transformers
- Security: cyber-physical systems security (attack graphs, fault trees, CVE-based analysis)
- Systems / Cloud: Docker, AWS (EC2, S3, Lambda), Google Cloud Platform, FastAPI, Flask, PostgreSQL, Git
- Also: Qiskit (quantum circuits, Grover's), ONNX Runtime, IoT (ESP32, MQTT)

MILITARY EXPERIENCE

National Service - Singapore Armed Forces

2020 - 2022

- Completed two years of full-time National Service, a mandatory commitment for Singaporean male citizens, serving in an operationally structured environment that demanded discipline, adaptability, and teamwork under pressure; a formative experience that continues to inform how I approach sustained, high-stakes technical work.

SOCIETAL EFFORTS

Live-in-Labs Field Immersion - Rural North India, Amrita Vishwa Vidyapeetham

- I spent an immersion period living in a rural community in North India through Amrita's Live-in-Labs programme, working directly with residents to understand their day-to-day challenges in education and healthcare, and co-designing practical interventions with them rather than for them.
- That immersion is where the idea for my mushroom cultivation digital twin project came from, a need I saw firsthand in the community. I've been building on it for over a year now (patent pending), and I plan to return to the same village to implement it.

Student Social Responsibility

2024

- Designed and taught STEM workshops for underprivileged students, aiming to broaden early exposure to computing and science for students who would otherwise have limited access to such resources.

HONOURS & SERVICE

- Head Prefect, Amrita Vidyalayam, Kollam (2018-2020) - led the student body and mentored junior students